

JECRC University, Jaipur
Department of Physics
B. Tech Ist Year – Ist Semester July-Dec 2025-26
Applied Physics (DPH001B)
Assignment-IV

- A. Very Short Answer Questions (10)**
1. What happens if the refractive index of core is kept lower than that of refractive index of cladding of an optical fiber? (2)
 2. Define fractional refractive Index for a fiber. (2)
 3. Write disadvantages of using Ruby LASER. (2)
 4. What is active medium used in semiconductor LASER? (2)
 5. Name the two steps involved in Holography? (2)
- B. Answer the following briefly (21)**
1. a) If the fractional refractive index of cladding with respect to core is 0.012 and numerical aperture of the optical fiber is 0.22, then find refractive indices of core and cladding of the optical fiber. (5)
b) What is digital holographic microscopy (DHM). (2)
 2. An optical fiber has refractive index of core to be 1.5 and relative refractive index difference of core and cladding to be 0.01. Determine numerical aperture and maximum angle of acceptance. (7)
 3. Using a schematic representation explain construction and working of semiconductor LASER. (7)
- C. Answer the followings explicitly. (33)**
1. Draw a well labeled diagram of an optical fiber. Carefully mark the various dimensions. Using a schematic diagram explain the construction of an optical fiber. Derive the formula for numerical aperture and maximum acceptance angle of an optical fiber. (11)
 2. What is Holography? Describe the process of construction and re-construction of a Hologram. Draw the necessary diagrams. (11)
 3. Using a schematic diagram explain the construction of He-Ne Laser. Explain its working using the energy level diagram. Write the applications of He-Ne Laser. (11)